Power Analysis: Introduction

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Introduction

Power analysis turns four quantities – effect size, significance level, power, and sample size – into a single equation: given any three, the fourth is determined. Every responsible study design pins down three of the four in advance to solve for the fourth (usually sample size). Skipping power analysis leaves studies under- or over-powered, with reputational and ethical consequences.

Prerequisites

Hypothesis testing, Type I and II errors.

Theory

The four quantities:

- **Effect size** (d, f, r, OR, RR): the magnitude of the true effect under the alternative. Must be pre-specified based on prior evidence, clinical relevance, or Cohen's conventions.
- **Significance level** α: the Type I error rate, almost always 0.05 two-sided.
- **Power** $1-\beta$: probability of detecting the true effect. Conventionally 0.80; confirmatory trials use 0.90.
- **Sample size** n: number of observations (or per group).

Given any three, the fourth is computed from the test's non-central distribution under H_1.

Sensitivity and trade-offs:

- Doubling n roughly shrinks the detectable effect by $\sqrt{2}$.
- Reducing α from 0.05 to 0.01 reduces power at the same n.
- Larger SD reduces power; more precise measurement increases it.
- Power for a one-sided test at α equals power for a two-sided test at 2α (when the effect is in the right direction).

Assumptions

The test's sampling distribution under H_1 must be known (or approximable). Power calculations are as valid as the assumed effect-size magnitude; a plausible range is often reported.

R Implementation

```r
library(pwr)

# Solve for n given d, alpha, power
pwr.t.test(d = 0.5, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")

# Solve for d given n and power (minimum detectable effect)
pwr.t.test(n = 30, sig.level = 0.05, power = 0.80,
           type = "two.sample")
```

```r
# Power curves across d
d_grid <- seq(0.1, 1, by = 0.05)
pw_n30  <- sapply(d_grid, function(d) pwr.t.test(n = 30, d = d)$power)
pw_n100 <- sapply(d_grid, function(d) pwr.t.test(n = 100, d = d)$power)

plot(d_grid, pw_n30, type = "l", col = "#F4A261", lwd = 2,
     xlab = "Cohen's d", ylab = "Power",
     main = "Power curves: n = 30 vs n = 100 per group")
lines(d_grid, pw_n100, col = "#2A9D8F", lwd = 2)
abline(h = 0.80, lty = 2)
legend("bottomright", c("n = 30", "n = 100"),
       col = c("#F4A261", "#2A9D8F"), lwd = 2)
```

Output & Results

```
     Two-sample t test power calculation

              n = 63.77
              d = 0.5
      sig.level = 0.05
          power = 0.80
    alternative = two.sided

     Two-sample t test power calculation

              n = 30
              d = 0.738
      sig.level = 0.05
          power = 0.80
```

64 per arm for a medium effect at 80 % power. With only 30 per arm, the minimum detectable effect is $d = 0.74$.

Interpretation

For a grant or protocol: "Assuming a between-group standardised difference of $d = 0.5$ (a medium effect), $\alpha = 0.05$ two-sided, and power = 0.80, the required sample size is 64 per arm (128 total). We plan to enrol 70 per arm to allow for 10 % loss to follow-up."

Practical Tips

- Choose the effect size from published data or pilot studies, not from Cohen's generic thresholds.
- Always inflate for dropout; report the final n as target.
- Sensitivity analysis: provide n for several plausible d values.
- Do not perform post-hoc power analysis with the observed effect; it is uninformative and sometimes misleading.
- For complex designs (cluster, multilevel), use simulation-based power (`simstudy`, custom Monte Carlo) rather than formulas.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**

- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation